

Capstone Project: Predicting Career Domain and Seniority from LinkedIn Profiles

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Agenda

- **Group Work**

- Data Exploratory
- Baseline Model
- Seniority Models
- Department Models

Group Work

Problem definition

Development of an end-to-end machine-learning pipeline that predicts (1) the current professional domain and (2) the current seniority level of an individual based solely

Data basis

- **In-Sample:** Training and Testing on provided data (Seniority.csv & Department.csv)
- **Out-of-Sample:** Profiles Dataset (LinkedIn Data)

Division of tasks

Leonhard Vogt

- Data exploratory
- Approach 3 & academic Research on existing models (JobBERT)
- Design and Structure of final report & presentation

Michael Back

- Data exploratory
- Approach 1 (Baseline exact matching + Extended rule based model)
- Approach 5 (Feature Engineering + Random Forest)

JinLian Chen

- Data exploratory
- Approach 7 (KNN model)
- Approach 2 (Zero shot)

Agenda

- Group Work
- **Data Exploratory**
- Baseline Model
- Seniority Models
- Department Models

EDA – All Datasets

Training data

LinkedIn data



Seniority.csv

count	
label	
Senior	3733
Lead	3546
Director	984
Management	756
Junior	409

≠

seniority	
Professional	216
Management	192
Lead	125
Senior	44
Director	34
Junior	12

- Class imbalance
- "Professional" is not in training data set
- Difference on class distribution between training data and Profiles Dataset (LinkedIn Data)

Department.csv

count	
label	
Marketing	4295
Sales	3328
Information Technology	1305
Business Development	620
Project Management	201
Consulting	167
Administrative	83
Other	42
Purchasing	40
Customer Support	33
Human Resources	31

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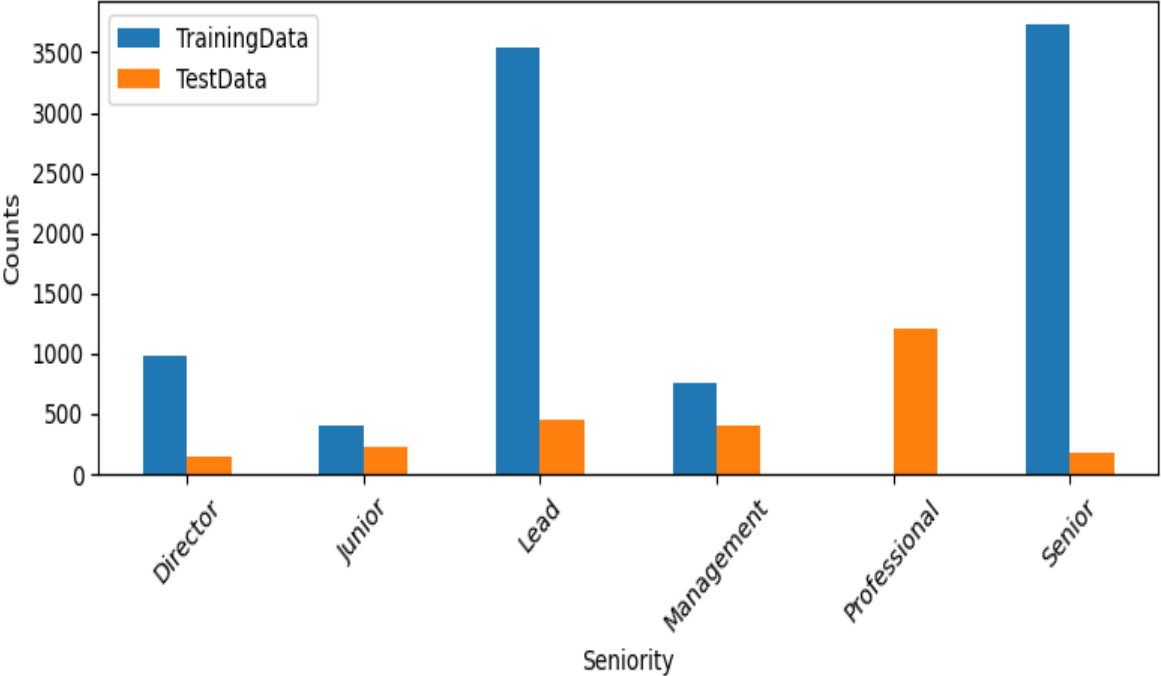
department	
Other	1252
Information Technology	312
Sales	219
Consulting	195
Project Management	175
Marketing	133
Administrative	84
Business Development	78
Purchasing	72
Human Resources	70
Customer Support	48

- Class imbalance
- “Other” is not a true class and is overrepresented in the Profiles Dataset (LinkedIn Data) but underrepresented in the training set
- In the training set, HR, Administrative, Customer Support, Purchasing, and “Other” are underrepresented in comparison and might therefore less usable for effective training

EDA – All Datasets

Seniority.csv

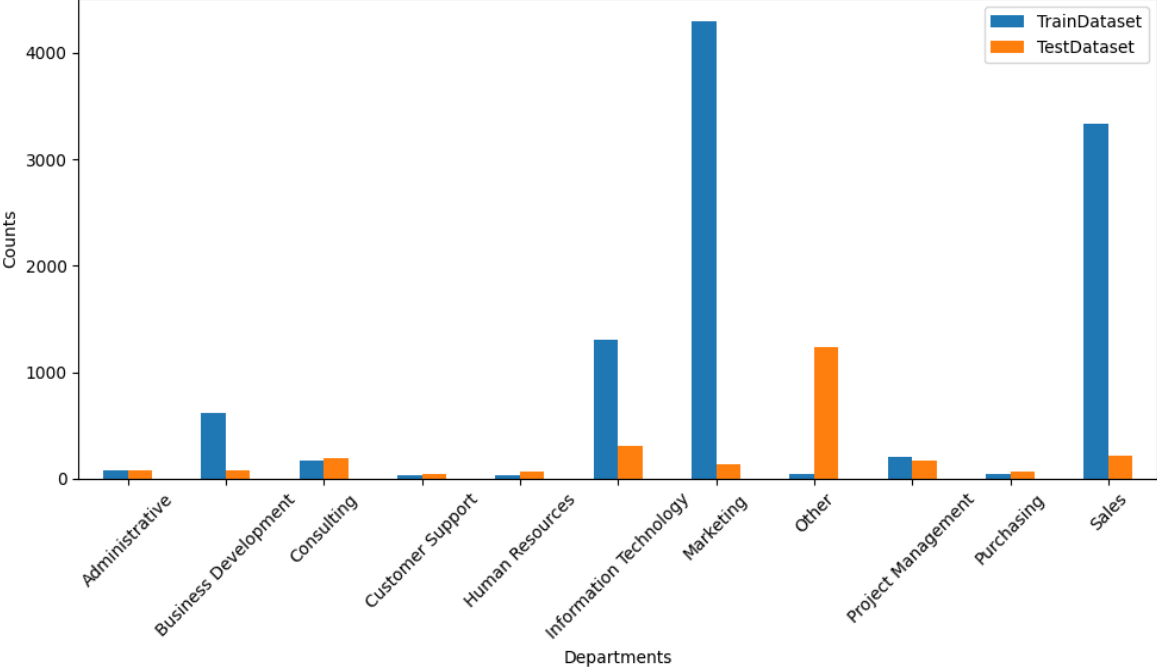
Seniority Counts Comparison



Source: <https://github.com/E-tech-coder/DataScienceCapstoneProject/blob/6014708a347b711ea84fac55ef01ec395be73235/SentenceTransformerKNN%20seniority.ipynb>

Department.csv

Department Counts Comparison



Source : <https://github.com/E-tech-coder/DataScienceCapstoneProject/blob/6014708a347b711ea84fac55ef01ec395be73235/Zero-shot%20prediction%20department.ipynb>

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Baseline Model – Rule-based matching

Approach 1

Rule-based matching (baseline)

Executive Summary

The rule-based exact matching baseline provides a transparent reference model with good accuracy on matched cases, confirming at most the correctness of the label lists. However, coverage on real-world LinkedIn CV data is very limited especially for department predictions resulting in low end-to-end accuracy mainly driven by *unknown* outputs rather than systematic errors. The model therefore defines a clear lower performance bound and highlights the need for more advanced rule-based or learning-based approaches.

Difficulties - Weakness

- High dependence on wording and spelling
- No ability to generalise
- Class imbalance between training and test set
- No ability to handle ambiguity

Mitigation strategies

- Text normalisation
- Separation of coverage and accuracy
- Use as baseline/reference model

Further improvements

Hybrid approach (Baseline + Fine-tuned)
Two-stage classification (Other vs. Specific)
More diverse training data
Firm- or time-based split

Coverage and end-to-end

Coverage seniority: 0.2187
Coverage department: 0.0627

- 157 out of 717 ACTIVE positions were matched for seniority
- 45 out of 717 ACTIVE positions were matched for department
- Highlights the limitation of this approach

Seniority end-to-end accuracy: 0.1546
Department end-to-end accuracy: 0.0585

- Limited due to only exact matching
- Serves as a baseline to be beaten

Extended Performance

Only matched cases (based on the 157 and 45 matched positions)

Seniority accuracy (matched only): 0.707
Matched seniority: 157

- Director and Senior -> high recall, low f1 score
- Junior low performance at all
- Management titles are sometimes misclassified as director or senior -> inconsistent assignment of seniority classes between training and test data

Department accuracy (matched only): 0.9333
Matched department: 45

- Performance decent overall
- Partial weakening in the reliability of forecasts for business development, IT and marketing
- Consulting has a lower recall

Baseline Model – Rule-based matching

Approach 1

Extended rule-based model (baseline)

Executive Summary

The extended rule-based model builds on the exact matching baseline by incorporating additional heuristic rules and normalized keyword patterns. This increases coverage and robustness while preserving interpretability. Although it still cannot fully generalize to unseen job titles, it represents a meaningful improvement over pure lookup-based matching and serves as an intermediate step between baseline and learning-based models.

Difficulties - Weakness

- Still rule- and keyword-dependent, no real semantics
- Manual maintenance of rules scales poorly
- Ambiguity (e.g. “consultant”, “associate”) can only be resolved to a limited extent
- Sensitive to incomplete or unusual titles

Mitigation strategies

- Text normalisation
- Advanced normalisation and keyword aggregation
- Hierarchical rules (e.g. specific before general)

Further improvements

- Further improvements of rule/heuristics are not the point and would lead to overengineering
- Use of more advanced, data driven approaches instead

Coverage and end-to-end

Seniority Prediction Coverage: 44.85%
Department Prediction Coverage: 23.68%

- Improvement for seniority from 22% to 45% -> over 200%
- Improvement for department from 6% to 23% -> nearly 400%

Seniority end-to-end accuracy: 36.77%
Department end-to-end accuracy: 20.89%

- Improvement for seniority from 15% to 36%
- Improvement for department from 6% to 20%
- But!
 - => Seniority: 38% of all profiles fall under the ‘professional’ category, which is not included in the training data
 - => Department: Over 56% of all cases are assigned to the ‘other’ category, which is not a clearly identifiable category

Performance - ‘other’ and ‘professional’ excluded

--- Seniority Evaluation
Accuracy: 59.19%

- Director -> lowest precision but highest recall
- Junior lowest performance at all -> semantics

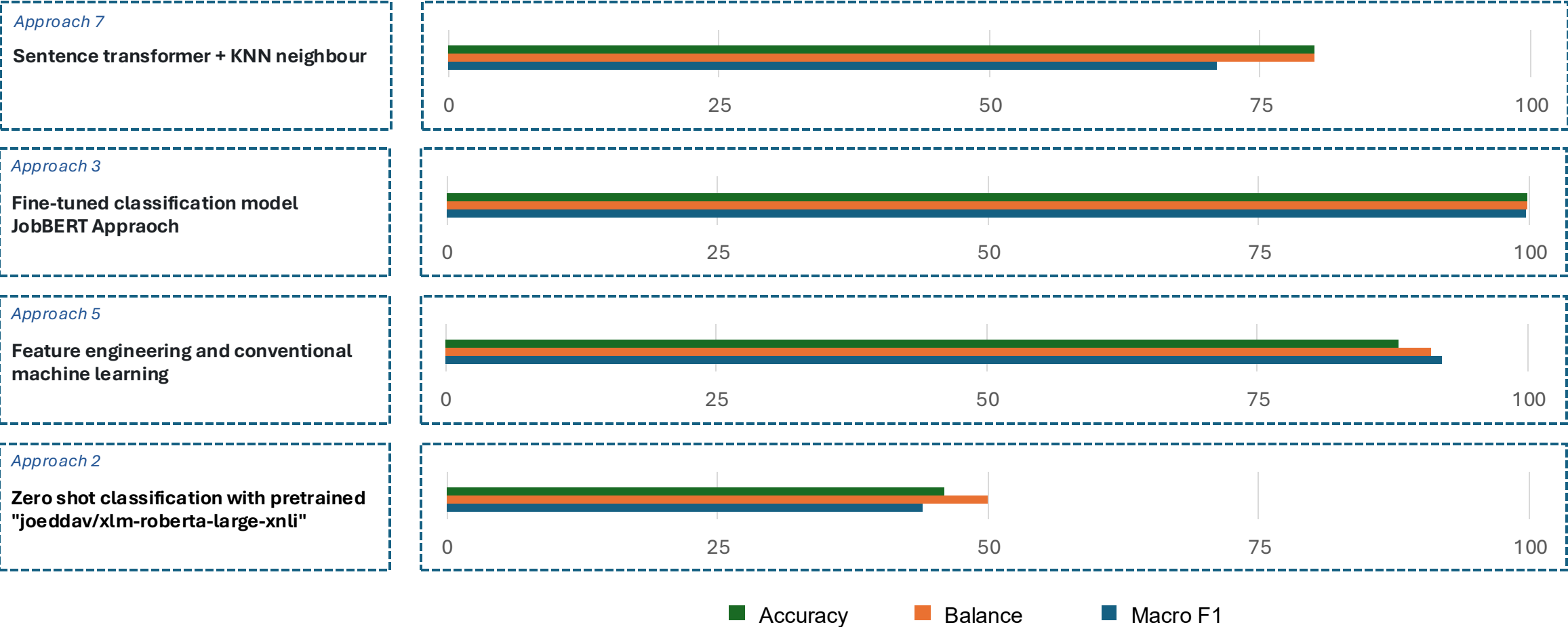
--- Department Evaluation
Accuracy: 47.92%

- Precision overall high, but low recall
- Macro avg precision 86%, recall 41%
 - => Many true cases are overlooked

Agenda

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- Department Models

Model Comparison – In-Sample Metrics for Seniority



Final Seniority Model – Fine Tuned Classification Model

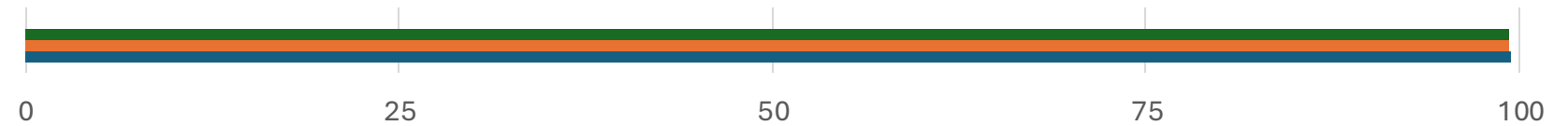
Underlying Data

**Baseline exact matching Out-of-Sample
Extended Baseline**



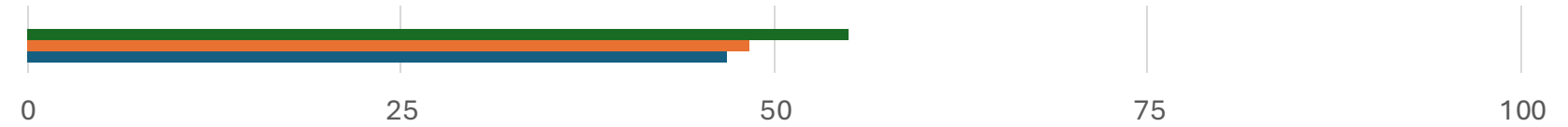
Underlying Data

Performance In-Sample



Underlying Data

Performance Out-of-Sample



■ Accuracy ■ Balance ■ Macro F1

Further Development

Finetuning on JobBERT is well-suited for seniority prediction as seniority-related signals are strongly embedded in job titles and descriptions. Through domain-specific pretraining, the model captures hierarchical and responsibility-related language patterns.

Fine-tuning aligns these latent representations with explicit seniority labels, enabling robust generalization to unseen profiles.

The “Professional” class poses a challenge due to its semantic ambiguity and overlap with adjacent seniority levels.

To address this, additional training data needs to be generated – training and testing (splitting by ID) indicates that **if the model gets trained on Professional, accuracy around 80 % could be achieved.**

Seniority Model – Fine-tuned classification model

Approach 3

Fine-tuned classification model JobBERT Approach

Difficulties - Weakness

- Computational extensive (GPU capacities for training & fine-tuning)
- Unforeseen class in LinkedIn Data

Mitigation strategies

- Solving not trained class by implementing a prediction confidence threshold for „professional“
- If Model isn't confident enough for one class, it receives the “professional” label

Further improvements

- Equally balanced training data
- Iterative fine-tuning to avoid concept drift over time
- Test Data Distribution

Executive Summary

Following a literature review, we identified prior research highlighting the relationship between job titles and their corresponding job descriptions. Building on these insights, we selected **JobBERT**, a domain-specific language model pre-trained on **21 million job-related data points**. In our approach, the Seniority CSV dataset was embedded using a **sentence transformer**. While additional fine-tuning led to improvements in in-sample accuracy, out-of-sample performance, indicates a **bias-variance trade-off between** our LogReg baseline and fine-tuned model.

In-Sample (Baseline on LogReg + JobBERT)

	precision	recall	f1-score	support
Director	0.882	0.944	0.912	197
Junior	0.738	0.927	0.822	82
Lead	0.912	0.911	0.912	709
Management	0.760	0.921	0.832	151
Senior	0.957	0.873	0.913	747
accuracy			0.901	1886
macro avg	0.850	0.915	0.878	1886
weighted avg	0.907	0.901	0.902	1886

In-Sample (Finetuned)

=== FINETUNED JobBERT (Seniority Test) ===				
	precision	recall	f1-score	support
Director	0.990	0.995	0.992	197
Junior	1.000	0.976	0.988	82
Lead	0.999	0.997	0.998	709
Management	0.993	0.987	0.990	151
Senior	0.993	0.997	0.995	747
accuracy			0.995	1886
macro avg	0.995	0.990	0.993	1886
weighted avg	0.995	0.995	0.995	1886

Out-of-Sample (with threshold for Professional)

- Reduced performance
- Untrained „Professional“ class significantly affects metrics
- Training with Professional indicates higher performance
- Mismatch between training data and Real world data
- Class imbalance

Accuracy	0.468
Macro F1	0.549
Balanced	0.483

Out-of-Sample (without Professional)

- For comparison purposes removing the untrained label allows for a realistic comparison of the OOS performance of the model
- Observable performance increase

Accuracy	0.711
Macro F1	0.608
Balanced	0.744

Department Model – Fine-tuned classification model

Approach 3

Fine-tuned classification model JobBERT Approach

Difficulties - Weakness

- Computational extensive (GPU capacities for training & fine-tuning)
- Label Logic for Annotated Profiles

Mitigation strategies

- Solving Class imbalance by implementing a prediction threshold for „other“
- Dumpster strategy for “other”
- If Model isnt confident enough for one class, it receives the “other“ label

Further improvements

- Equally balanced training data
- Iterative fine-tuning to avoid concept drift over time
- Test Data Distribution

Executive Summary

Following a literature review, we identified prior research highlighting the relationship between job titles and their corresponding job descriptions. Building on these insights, we selected **JobBERT**, a domain-specific language model pre-trained on **21 million job-related data points**. In our approach, the Department CSV dataset was embedded using a **sentence transformer**. While fine-tuning led to improvements for in-sample accuracy, out-of-sample performance, indicates a **bias–variance trade-off between** our LogReg baseline and fine-tuned model.

In-Sample ((Baseline on LogReg + JobBERT)

```
=== BASELINE: Embeddings + LogisticRegression (Department Test) ===
Accuracy: 0.9295219319862001
Macro F1: 0.8383296735008721
```

	precision	recall	f1-score	support
Administrative	0.519	0.824	0.636	17
Business Development	0.776	0.919	0.841	124
Consulting	0.780	0.970	0.865	33
Customer Support	0.750	0.857	0.800	7
Human Resources	0.600	1.000	0.750	6
Information Technology	0.890	0.897	0.893	261
Marketing	0.984	0.936	0.959	859
Other	0.727	1.000	0.842	8
Project Management	0.636	0.875	0.737	40
Purchasing	0.889	1.000	0.941	8
Sales	0.975	0.938	0.956	666
accuracy			0.930	2029
macro avg	0.775	0.929	0.838	2029
weighted avg	0.939	0.930	0.932	2029

In-Sample (Finetuned)

```
=== FINETUNED JobBERT (Department Test) ===
```

	precision	recall	f1-score	support
Administrative	1.000	1.000	1.000	17
Business Development	0.992	0.992	0.992	124
Consulting	1.000	1.000	1.000	33
Customer Support	1.000	1.000	1.000	7
Human Resources	1.000	1.000	1.000	6
Information Technology	0.992	0.989	0.990	261
Marketing	1.000	0.999	0.999	859
Other	1.000	1.000	1.000	8
Project Management	1.000	1.000	1.000	40
Purchasing	1.000	1.000	1.000	8
Sales	0.997	1.000	0.999	666
accuracy			0.998	2029
macro avg	0.998	0.998	0.998	2029
weighted avg	0.998	0.998	0.998	2029

Out-of-Sample (with threshold for Other)

- Substantially lower than the in-sample performance
- Does not generalise well (due to unprecedented data)
- Mismatch between training data and Real world data
- Class imbalance

Accuracy	0.553
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Macro F1	0.407
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Balanced	0.550
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Out-of-Sample (without “Other”)

- Solid but still does not generalise very well
- The precision for Administrative, Consulting, Customer Support and Project Management is very high, but recall is quite low -> correct but incomplete
- For IT, it's exactly the opposite -> low precision, high recall
- Sales, Marketing and Business Development are mid but balanced
- Purchasing was not correctly identified at all (class imbalance)

Accuracy	0.683
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Macro F1	0.597
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Balanced	0.687
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Department Model – Feature engineering + conventional machine learning

Approach 5

Feature engineering and conventional machine learning

Executive Summary

The feature engineering and embedding-based Random Forest model leverages semantic representations of job titles to generalize beyond exact matching. Compared to the extended rule-based approach, it achieves approximately 10% higher end-to-end performance on real-world LinkedIn CV data, indicating improved generalization for department prediction, while still facing challenges due to semantic overlap and ambiguity between departments.

Difficulties - Weakness

- Class imbalance affects recall
- Label noise and heterogeneous job titles limit upper performance for this approach
- Performance depends on embedding quality and feature design

Mitigation strategies

- Text normalization + feature engineering before embedding
- Evaluation with precision, recall and F1 instead of accuracy only
- Separation of train, validation and out-of-sample test sets

Further improvements

- Expansion and balancing of training data
- Advanced embeddings (domain-specific)
- Threshold-based confidence filtering or top-k predictions
- Ensemble approaches or calibration of predicted probabilities

Model Architecture (high Level)

(High-Level)

- Text Normalization
- Feature Engineering
 - Structure features
 - Numerical keyword features
 - Multilingual word embeddings
- Random Forest Classifier

In-Sample

Accuracy: 0.9522

Classification Report:

	precision	recall	f1-score	support
Administrative	0.75	0.88	0.81	17
Business Development	0.96	0.91	0.93	124
Consulting	0.91	0.88	0.89	33
Customer Support	1.00	0.29	0.44	7
Human Resources	1.00	0.67	0.80	6
Information Technology	0.84	0.97	0.90	261
Marketing	0.99	0.98	0.98	859
Other	1.00	0.38	0.55	8
Project Management	0.79	0.78	0.78	40
Purchasing	1.00	0.38	0.55	8
Sales	0.97	0.96	0.97	666
accuracy			0.95	2029
macro avg	0.93	0.73	0.78	2029
weighted avg	0.96	0.95	0.95	2029

Out-of-Sample

- Substantially lower than the in-sample performance
- Does not generalise well (due to unprecedented data)
- Mismatch between training data and Real world data
- Class imbalance

Accuracy	0.25
Macro F1	0.37
Balanced	0.19

Out-of-Sample (without "Other")

- Solid but still does not generalise very well
- The precision for Administrative, Consulting, Customer Support and Project Management is very high, but recall is quite low
-> correct but incomplete
- For IT, it's exactly the opposite
-> low precision, high recall
- Sales, Marketing and Business Development are mid but balanced
- Purchasing was not correctly identified at all (class imbalance)

Accuracy	0.57
Macro F1	0.45
Balanced	0.55

Department Model - Zero shot classification model

Approach 2

Fine tune the Zero shot classification model "joeddav/xlm-roberta-large-xnli" with our department dataset

Executive Summary

Xlm-roberta-large-xnli is not a standard classifier. It is trained for Natural Language Inference (NLI): premise + hypothesis → {entailment, neutral, contradiction}. Zero-shot classification works because Hugging Face reformulates classification as NLI. Fine-tuning this model directly with "position"- "department" is not appropriate. Its strength lies in NLI-based zero-shot inference.

Difficulties - Weakness

1. The training data format "premis" - "hypothesis" of NLI model is different from the "position"- "department" pattern in our task.
2. The original model has 3 labels , but we have 11 department labels.

Mitigation strategies

A model pretrained for sequence classification.
Or reformulating training data explicitly as NLI pairs. (Not appropriate for department prediction.)

Further improvements

Choose models specifically trained on short job titles.

Model Architecture

- Label encoding
- Prepare training(department.csv) and evaluation (Linkedin) dataset
- Tokenization
- Training and evalution

Out-of-sample

Data: Profiles Dataset (LinkedIn Data)
After-training Accuracy : 12.8%

```
[1269/1269 03:35]
{'eval_loss': 2.3762354850769043,
 'eval_model_preparation_time': 0.0689,
 'eval_accuracy': 0.12863479546574666,
 'eval_runtime': 218.3421,
 'eval_samples_per_second': 46.464,
 'eval_steps_per_second': 5.812}
```

In-Sample

Data: "Department.csv"

Accuracy	0.74
Macro F1	0.47
Balanced	0.77

Out-of-Sample

Data: Profiles Dataset (LinkedIn Data)
Before training Accuracy : 56%

	precision	recall	f1-score	support
Administrative	0.15	0.54	0.23	84
Business Development	0.49	0.51	0.50	78
Consulting	0.46	0.62	0.53	195
Customer Support	0.62	0.60	0.61	48
Human Resources	0.30	0.91	0.45	69
Information Technology	0.69	0.63	0.66	309
Marketing	0.58	0.40	0.47	133
Other	0.73	0.53	0.61	1235
Project Management	0.74	0.68	0.70	173
Purchasing	0.32	0.60	0.42	72
Sales	0.79	0.51	0.62	219
accuracy			0.56	2615
macro avg	0.53	0.59	0.53	2615
weighted avg	0.65	0.56	0.59	2615

Agenda

- Group Work
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- **Department Models**

Model Comparison – In-Sample Metrics for Department



Department Model – Sentence transformer + KNN neighbour

Approach 7

Sentence transformer from TechWolf/JobBERT-v3 + KNN distance threshold

Difficulties - Weakness

Label distribution shift. In our training data, the KNN model didn't get enough semantic training on the "Other" department, while in the test dataset half of the data is labeled as "Other".

Mitigation strategies

- Treat "Other" department as a spare category instead of a normal label. Assign "Other" when the evaluation sample is further than the threshold value from other training samples.

Further improvements

- Compare sample data with real world data, be careful of class imbalance during data collection.

Executive Summary

- Label distribution shift in the training and evaluation dataset and class imbalance can hurt the model performance since the model face a new pattern that it is not familiar with in the evalution process.
- We need to consider it during data collection and analysis, or find a special treatment for the new labels, or use different evaluation criteria.

Model Architecture

- Create embeddings with the sentence transformer from "TechWolf/JobBERT-v3"
- Train KNN Classifier with "Deaprtment.csv" dataset
- Evaluate the test data and calculate this distances
- Replace the prediction with "Other" department when similarity is small with the training data
- Measure the accuracy on the CV dataset

Out-of-Sample (Without distance threshold)

Data: Profiles Dataset (LinkedIn Data)
Accuracy : 38% (K = 15)

	precision	recall	f1-score	support
Administrative	0.30	0.39	0.34	84
Business Development	0.25	0.82	0.38	78
Consulting	0.75	0.50	0.60	195
Customer Support	0.88	0.15	0.25	48
Human Resources	0.82	0.68	0.75	69
Information Technology	0.65	0.67	0.66	309
Marketing	0.15	0.88	0.26	133
Other	0.94	0.05	0.10	1235
Project Management	0.49	0.58	0.53	173
Purchasing	0.76	0.74	0.75	72
Sales	0.31	0.89	0.46	219
accuracy			0.38	2615
macro avg	0.57	0.58	0.46	2615
weighted avg	0.72	0.38	0.33	2615

In sample

Data: Department.csv

Accuracy	0.93
Macro F1	0.73
Balanced	0.94

Out-of-Sample (With distance threshold)

Data: Profiles Dataset (LinkedIn Data)
Accuracy : 63.3% (K=15)

Threshold =0.711, Accuracy=0.500
Threshold =0.737, Accuracy=0.590
Threshold =0.763, Accuracy=0.603
Threshold =0.789, Accuracy=0.631
Threshold =0.816, Accuracy=0.633
Threshold =0.842, Accuracy=0.628
Threshold =0.868, Accuracy=0.617
Threshold =0.895, Accuracy=0.608

Final Department Model – Sentence transformer + KNN neighbour

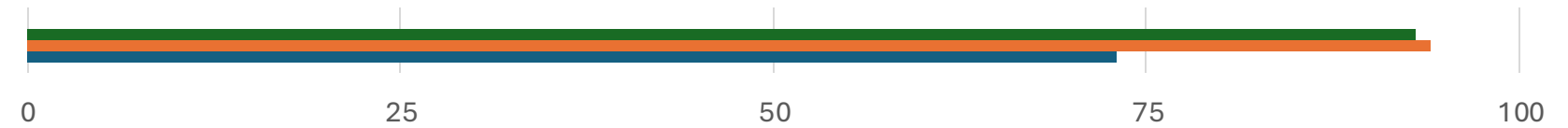
Underlying Data

**Baseline exact matching Out-of-Sample
Extended Baseline**



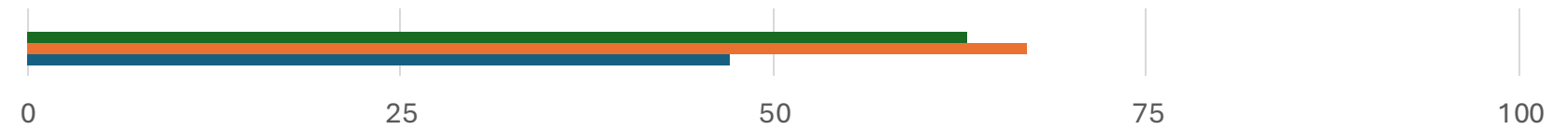
Underlying Data

Performance In-Sample



Underlying Data

Performance Out-of-Sample



■ Accuracy ■ Balance ■ Macro F1

Further Development

- Data Quality and Distribution Analysis
Conduct a systematic comparison between training data and real-world job title distributions to identify and mitigate class imbalance and distribution shift during data collection.
- Different similarity threshold for each class for better accuracy
- Alternative Classification Strategies
Evaluate supervised classifiers (e.g. SVM, gradient boosting) on top of sentence embeddings for performance comparison.
- Robust Evaluation Metrics
Extend evaluation beyond overall accuracy by incorporating macro-F1 scores, per-class recall, and out-of-distribution detection performance to better reflect real-world model behavior.

Seniority Model – sentence transformer + KNN neighbour

Approach 7

Sentence transformer from TechWolf/JobBERT-v3 + KNN distance threshold

Difficulties - Weakness

Label distribution shift. In our training data there's a large proportion of "Lead" and "Senior" labeled data. But in the evaluation set, we have mostly "Professional" labels that are not included in the training data.

Mitigation strategies

- Distance threshold here doesn't help with accuracy because of the data.
- The strategy here is to reject the labels not appearing in the training dataset, in order to give a fair review on our model.

Further improvements

- Compare sample data with real world data, be careful of class imbalance during data collection.
- Use other evaluation methods

Executive Summary

With our distance threshold method, if the sample is too far from our training data, we dump it to the "Professional" label. It gives us a highest accuracy of 33.7%, which is the same as the accuracy before we apply this distance threshold method. Distance threshold method doesn't help. Possible reason: Label ambiguity dominates the task. LinkedIn titles like: "Associate" "Consultant" "Staff" "Member" "Analyst" are: semantically very close, but map to different seniority labels So: high cosine similarity $\neq$ clear label

Model Architecture

- Create embeddings with the sentence transformer from "TechWolf/JobBERT-v3"
- Train KNN Classifier with "Seniority.csv" dataset
- Measure the accuracy on the CV dataset
- Measure the accuracy on the CV dataset filtering out "Professional" label

In-Sample

Data: Seniority.csv

Accuracy : 79% (K = 21)

	precision	recall	f1-score	support
Director	0.56	0.81	0.66	146
Junior	0.46	0.78	0.58	55
Lead	0.87	0.77	0.82	802
Management	0.67	0.64	0.65	148
Senior	0.84	0.84	0.84	735
accuracy			0.79	1886
macro avg	0.68	0.77	0.71	1886
weighted avg	0.81	0.79	0.80	1886

Out-of-Sample

Data: Profiles Dataset (LinkedIn Data)

Accuracy : 34% (K = 21)

	precision	recall	f1-score	support
Director	0.45	0.46	0.45	136
Junior	0.41	0.43	0.42	219
Lead	0.71	0.40	0.51	801
Management	0.68	0.74	0.71	374
Professional	0.00	0.00	0.00	0
Senior	0.74	0.12	0.20	1085
accuracy			0.34	2615
macro avg	0.50	0.36	0.38	2615
weighted avg	0.68	0.34	0.40	2615

Out-of-Sample (Filtering out "Professional")

Data: Profiles Dataset (LinkedIn Data)

Accuracy : 64% (K = 21)

	precision	recall	f1-score	support
Director	0.43	0.48	0.45	124
Junior	0.47	0.90	0.62	121
Lead	0.71	0.68	0.70	473
Management	0.71	0.83	0.76	347
Senior	0.71	0.35	0.47	339
accuracy			0.64	1404
macro avg	0.60	0.65	0.60	1404
weighted avg	0.66	0.64	0.63	1404

Seniority Model – Feature engineering and conventional machine learning

Approach 5

Feature engineering and conventional machine learning

Executive Summary

The feature engineering and embedding-based Random Forest model leverages semantic representations of job titles to generalize beyond exact matches. Compared to rule-based approaches, it achieves substantially higher coverage and better end-to-end performance on real-world LinkedIn CV data, while still maintaining reasonable interpretability through feature importance and similarity-based explanations.

Difficulties - Weakness

- Class imbalance affects recall
- Label noise and heterogeneous job titles limit upper performance for this approach
- Performance depends on embedding quality and feature design

Mitigation strategies

- Text normalization + feature engineering before embedding
- Evaluation with precision, recall and F1 instead of accuracy only
- Separation of train, validation and out-of-sample test sets

Further improvements

- Expansion and balancing of training data
- Advanced embeddings (domain-specific)
- Threshold-based confidence filtering or top-k predictions
- Ensemble approaches or calibration of predicted probabilities

Model Architecture (high Level)

(High-Level)

- Text Normalization
- Feature Engineering
 - Structure features
 - Numerical keyword features
 - Multilingual word embeddings
- Random Forest Classifier

In-Sample

Accuracy: 0.9168

Classification Report:

	precision	recall	f1-score	support
Director	0.99	0.97	0.98	197
Junior	1.00	0.54	0.70	82
Lead	0.89	0.95	0.92	709
Management	0.95	0.84	0.89	151
Senior	0.92	0.93	0.92	747
accuracy			0.92	1886
macro avg	0.95	0.85	0.88	1886
weighted avg	0.92	0.92	0.91	1886

Out-of-Sample

- Substantially lower than the in-sample performance
- Does not generalize well (due to unprecedented data)
- Mismatch between training data and Real world data
- Class imbalance

Accuracy	0.41
Macro F1	0.41
Balanced	0.36

Out-of-Sample (without Professional)

- Solid but still does not generalize very well
- The precision for junior is very high, but recall is quite low -> correct but incomplete.
- For senior, it's exactly the opposite -> low precision, high recall
- Director, Lead and Management are decent

Accuracy	0.66
Macro F1	0.59
Balanced	0.68

Seniority Model - Zero shot classification model

Approach 2

Zero shot classification model
"joeddav/xlm-roberta-large-xnli"

Executive Summary

Using job title alone to predict seniority with only zero shot model is not an ideal solution. Since job title provides very limited information for the seniority. We can try feeding more information combined with the job title to predict the seniority or combine different models for a more flexible prediction.

Difficulties - Weakness

Seniority terms (e.g., Junior, Lead, Manager) are more subtle and often ambiguous in job titles. Zero-shot models struggle when label distinctions are not explicit in text.

Mitigation strategies

Data engineering – add context in the job title, for example, experiences, responsibilities.

Further improvements

Combine rule-based model with zero shot model. For example:
IF title contains ["Head", "Director", "VP"] → Senior
ELIF title contains ["Intern", "Trainee", "Junior"] → Junior
ELSE → Zero-shot model

Model Architecture

- Import model pipeline
- Zero-shot prediction on department with a pretrained model based on position name
- Evaluation model accuracy with classification report

In-Sample

Data: "Seniority.csv"
Accuracy : 46%

	precision	recall	f1-score	support
Director	0.56	0.86	0.68	984
Junior	0.75	0.28	0.41	409
Lead	0.81	0.52	0.64	3546
Management	0.04	0.11	0.06	756
Senior	0.44	0.39	0.42	3733
accuracy			0.46	9428
macro avg	0.52	0.43	0.44	9428
weighted avg	0.58	0.46	0.50	9428

Out-of-sample

Data: Profiles Dataset (LinkedIn Data)

Accuracy : 52 %

	precision	recall	f1-score	support
Director	0.48	0.78	0.60	141
Junior	0.84	0.54	0.66	230
Lead	0.54	0.29	0.38	455
Management	0.05	0.03	0.04	408
Professional	0.71	0.73	0.72	1211
Senior	0.16	0.52	0.25	170
accuracy			0.52	2615
macro avg	0.47	0.48	0.44	2615
weighted avg	0.54	0.52	0.51	2615

Out-of-Sample (Filtering out "Professional")

Data: Profiles Dataset (LinkedIn Data)

Accuracy : 43% (K =21)

	precision	recall	f1-score	support
Director	0.48	0.79	0.60	141
Junior	0.97	0.57	0.71	230
Lead	0.67	0.38	0.49	455
Management	0.19	0.05	0.08	408
Senior	0.25	0.96	0.39	170
accuracy			0.43	1404
macro avg	0.51	0.55	0.45	1404
weighted avg	0.51	0.43	0.41	1404

Thank you for listening

Declaration on usage of Generative AI

Group Member

Leonhard Vogt

Usage of Generative AI

- Starting Point for Idea Generation
- Supervised code generation for Models by providing context code from course lectures
- Debugging Code via Colab AI Plugin

Group Member

Michael Back

- Ideas and structural support
- Code generation beyond lecture scope
- Code explanation, debugging and error resolution
- Conceptual clarification and model interpretation

Group Member

JinLian Chen

- Model improvement
- Code explanation, debugging and error resolution
- Help on model performance interpretation
- Assistance in the functions of model parameters